Lab Sheet: 3

Objectives

- **To understand the use of COMMIT and ROLLBACK commands in transaction control.**
- **To use ORDER BY and GROUP BY clauses in SQL queries.**
- **To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.**
- **To understand and implement JOIN operations in SQL.**

Instructions

Open your preferred Database Management System (DBMS) tool.

1. **Create the required tables: TEACHER, SALES, DEPARTMENTS, and EMPLOYEES.**
2. **Insert suitable records into each table.**
3. **Execute the SQL commands for transaction control, privilege control, grouping, sorting, and joins.**
4. **Observe the output after each operation such as ROLLBACK, COMMIT, SAVEPOINT, GRANT, and REVOKE.**

DDL Questions (Data Definition Language)

1. Create the **TEACHER** table with the following attributes:
(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
CREATE TABLE TEACHER (
  TeacherID INT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(50) NOT NULL,
  DeptNo INT,
  Date_of_Joining DATE,
  DeptName VARCHAR(50),
```

```
  Location VARCHAR(50),
  Salary DECIMAL(10, 2)
);
```

2. Create the **SALES** table with the following attributes:
(Sales_No, Salesname, Branch, Salesamount, DOB).

```
CREATE TABLE SALES (
  Sales_No INT PRIMARY KEY,
  Salesname VARCHAR(50),
  Branch VARCHAR(50),
  Salesamount DECIMAL(12, 2),
  DOB DATE
);
```

3. Create the **DEPARTMENTS** table with the following attributes:
(dept_no, dept_name, dept_location).

```
CREATE TABLE DEPARTMENTS (
  dept_no INT PRIMARY KEY,
  dept_name VARCHAR(50) NOT NULL,
  dept_location VARCHAR(50)
);
```

4. Create the **EMPLOYEES** table with the following attributes:
(emp_id, emp_name, emp_salary, dept_no).

```
CREATE TABLE EMPLOYEES (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50) NOT NULL,
  emp_salary DECIMAL(10, 2),
  dept_no INT,
  FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

DML Questions (Data Manipulation Language)

From TEACHER Table

1. Insert five records into the **TEACHER** table.

INSERT INTO TEACHER (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary) VALUES

('Dr. Sharma', 10, '2020-01-15', 'Mathematics', 'Block A', 60000),

('Prof. Gupta', 20, '2019-03-22', 'Commerce', 'Block B', 55000),

('Mr. Khan', 10, '2021-08-10', 'Mathematics', 'Block A', 50000),

('Ms. Iyer', 20, '2018-11-05', 'Commerce', 'Block B', 58000),

('Dr. Reddy', 30, '2022-05-12', 'Science', 'Block C', 65000);

2. Give an increment of 25% salary to teachers working in the Mathematics department.

UPDATE TEACHER

SET Salary = Salary * 1.25

WHERE DeptName = 'Mathematics';

3. Give an increment of 15% salary to teachers working in the Commerce department.

UPDATE TEACHER

SET Salary = Salary * 1.15

WHERE DeptName = 'Commerce';

From SALES Table

1. Insert five records into the **SALES** table.

INSERT INTO SALES (Sales_No, Salesname, Branch, Salesamount, DOB) VALUES

(101, 'Rahul', 'North', 50000, '1995-12-21'),

(102, 'Sonia', 'East', 75000, '1992-05-14'),

(103, 'Amit', 'North', 40000, '1998-12-09'),

(104, 'Vikram', 'West', 60000, '1994-01-25'),

(105, 'Anjali', 'East', 80000, '1993-12-30');

2. Calculate the total sales amount in each branch.

SELECT Branch, SUM(Salesamount) AS Total_Sales

FROM SALES

GROUP BY Branch;

3. Calculate the average sales amount in each branch.

SELECT Branch, AVG(Salesamount) AS Avg_Sales

FROM SALES

GROUP BY Branch;

4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

SELECT Salesname, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB

FROM SALES

WHERE MONTH(DOB) = 12;

Salesname	DOB
Rahul	21-Dec-95
Amit	09-Dec-98
Anjali	30-Dec-93

5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

SELECT Salesname, DOB

FROM SALES

ORDER BY DATE_FORMAT(DOB, '%M');

Salesname	DOB
Rahul	1995-12-21
Amit	1998-12-09
Anjali	1993-12-30
Vikram	1994-01-25
Sonia	1992-05-14

From JOIN Operations

1. Display employee details and department details where the department numbers are the same in both tables.

SELECT E.*, D.* FROM EMPLOYEES E

INNER JOIN DEPARTMENTS D ON E.dept_no = D.dept_no;

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
501	Aman Verma	45000.00	10	10	Human Resources	Delhi
502	Sonal Singh	62000.00	20	20	Information Technology	Bangalore
503	Chris Evans	58000.00	20	20	Information Technology	Bangalore
504	Riya Kapoor	51000.00	30	30	Finance	Mumbai
505	Kabir Khan	49000.00	40	40	Marketing	Pune

2. Display the employee name and department name using a LEFT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name

FROM EMPLOYEES E

LEFT JOIN DEPARTMENTS D ON E.dept_no = D.dept_no;
```

emp_name	dept_name
Aman Verma	Human Resources
Sonal Singh	Information Technology
Chris Evans	Information Technology
Riya Kapoor	Finance
Kabir Khan	Marketing

3. Display the employee name and department name using a RIGHT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name

FROM EMPLOYEES E

RIGHT JOIN DEPARTMENTS D ON E.dept_no = D.dept_no;
```

emp_name	dept_name
Aman Verma	Human Resources
Sonal Singh	Information Technology
Chris Evans	Information Technology
Riya Kapoor	Finance
Kabir Khan	Marketing
NULL	Sales

4. Display the details of employees who draw a salary greater than the average salary.

```
SELECT * FROM EMPLOYEES

WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

emp_id	emp_name	emp_salary	dept_no
502	Sonal Singh	62000.00	20
503	Chris Evans	58000.00	20
NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1. Perform the **ROLLBACK** command on the TEACHER table.

UPDATE TEACHER SET Salary = Salary * 5;

ROLLBACK TO SAVEPOINT SP1;

ROLLBACK;

2. Perform the **COMMIT** command on the TEACHER table.

COMMIT;

3. Write a query to implement a **SAVEPOINT**.

START TRANSACTION;

INSERT INTO TEACHER (Name, DeptName, Salary)

VALUES ('New Instructor', 'Physics', 45000);

SAVEPOINT SP1;

DCL Questions (Data Control Language)

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.

GRANT ALL PRIVILEGES ON EMPLOYEES TO 'DEPARTMENTS'@'localhost';

2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.

GRANT SELECT, UPDATE ON EMPLOYEES TO 'DEPARTMENTS'@'localhost';

3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.

REVOKE ALL PRIVILEGES ON EMPLOYEES FROM 'DEPARTMENTS'@'localhost';

4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

REVOKE UPDATE ON EMPLOYEES FROM 'DEPARTMENTS'@'localhost';